Module 19

Entity-Relationship Model/2

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Module Recap

- Design Process for Database Systems

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- Design Process for Database Systems
- ER Model for real world representation with entities, entity sets, attributes, and relationships

Module Objectives

Goals

- To illustrate ER Diagram notation for ER Models

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- To illustrate ER Diagram notation for ER Models
- To explore translation of ER Models to Relational Schemas

Module Outline

- ER Diagram

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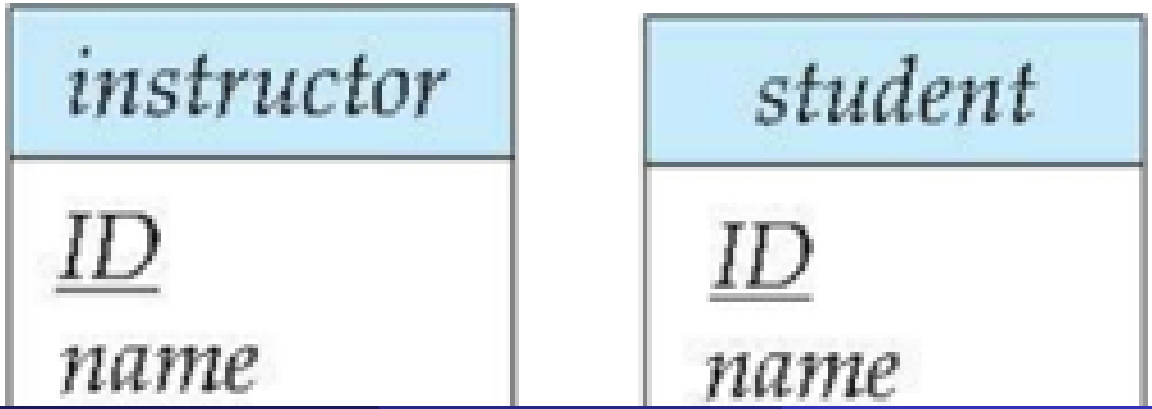
- ER Model to Relational Schema

ER Diagram

Entity Sets

Entities can be represented graphically as follows:

- Rectangles represent entity sets.



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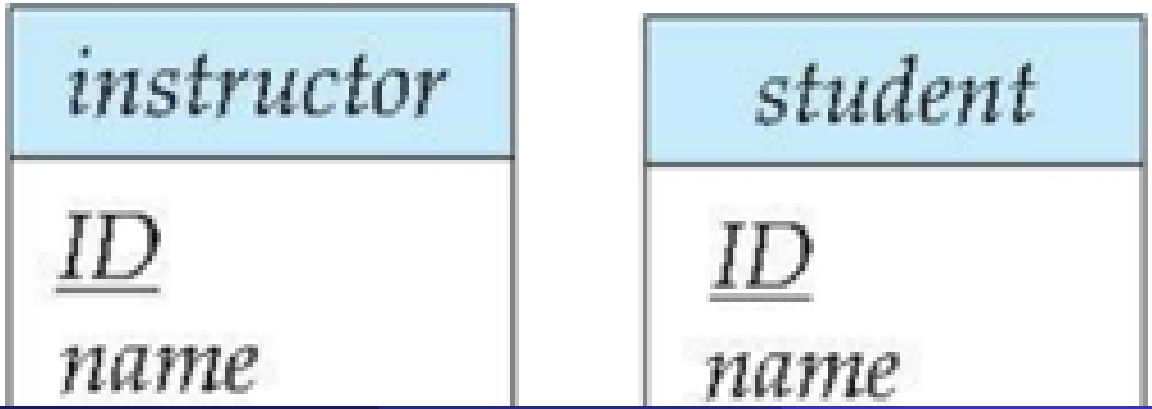

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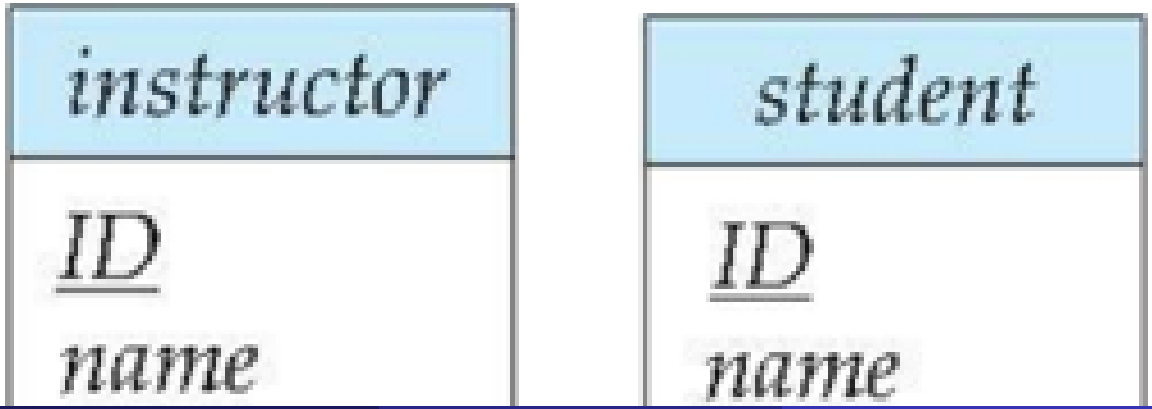
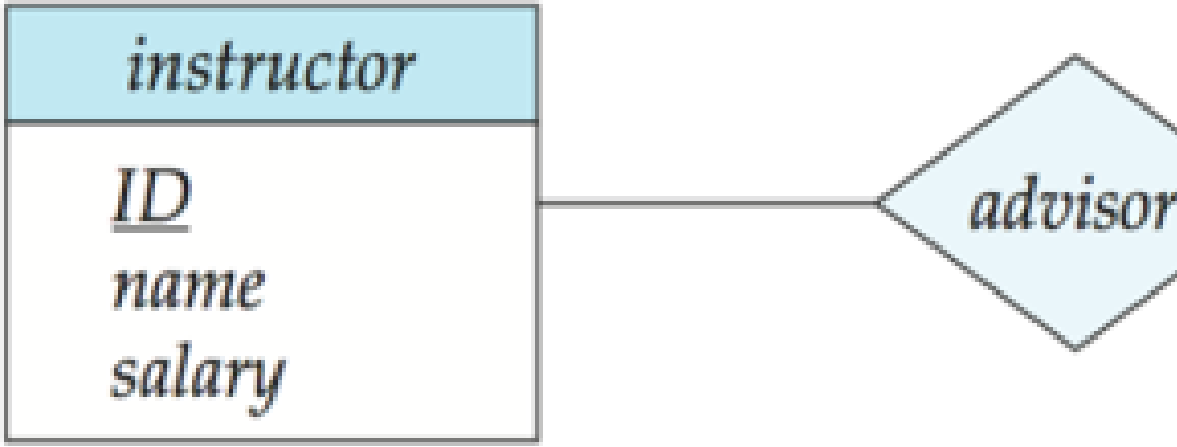
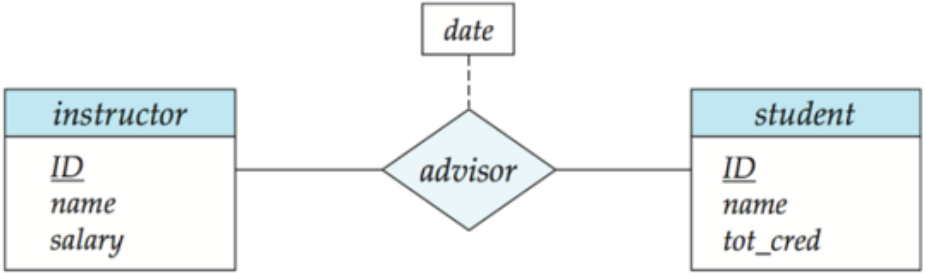


Relationship Sets

- Diamonds represent relationship sets.



Relationship Sets with Attributes



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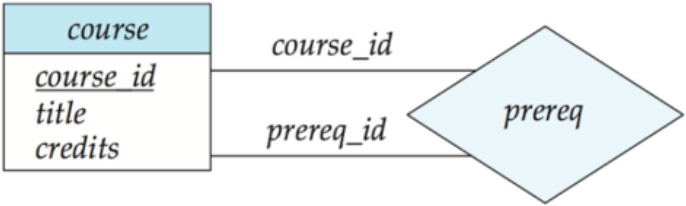

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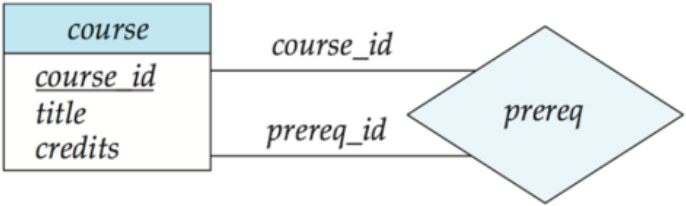
- The labels 'course id' and 'prereq id' are called **roles**.



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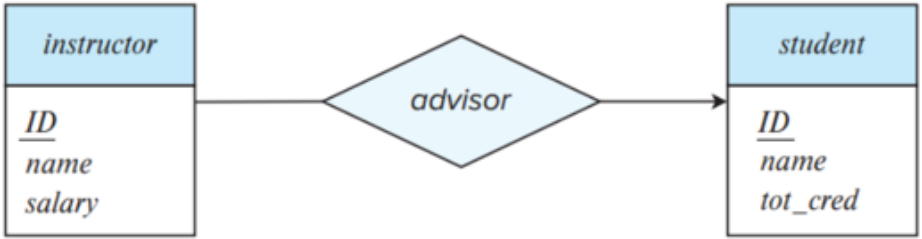

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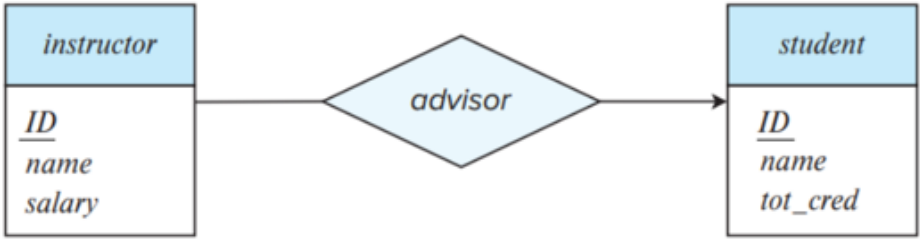

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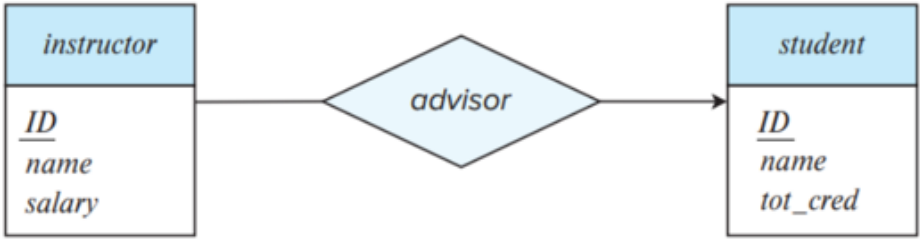

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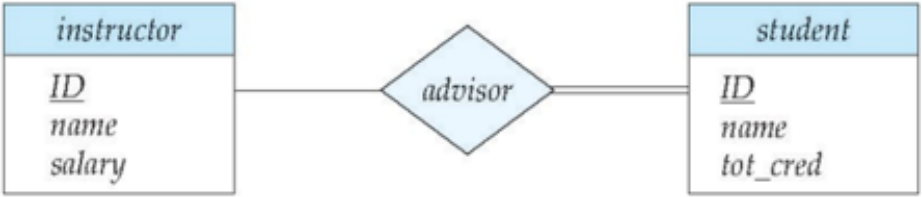

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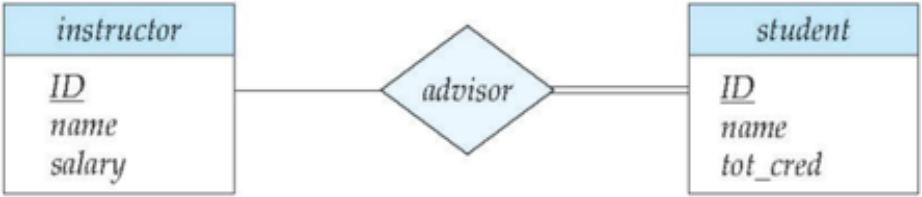

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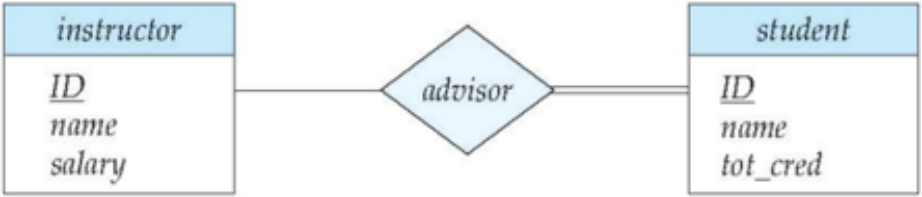

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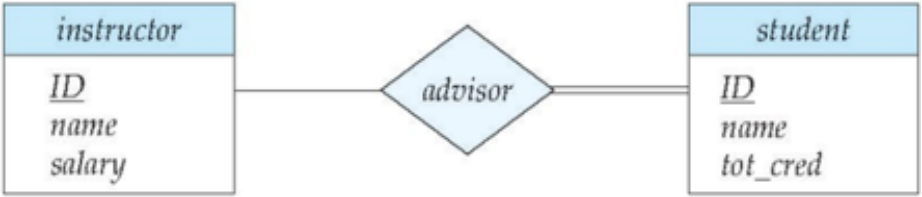

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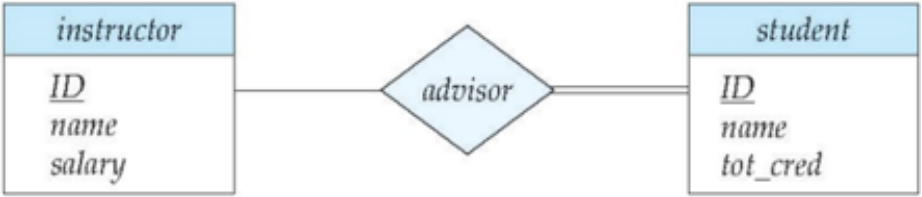
  - Example: participation of instructor in advisor is partial



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- A maximum value of * indicates no limit.

Notation to Express Entity with Complex Attributes

instructor

- *ID*
- *name*
  - *first_name*
  - *middle_initial*
  - *last_name*
- *address*
  - *street*

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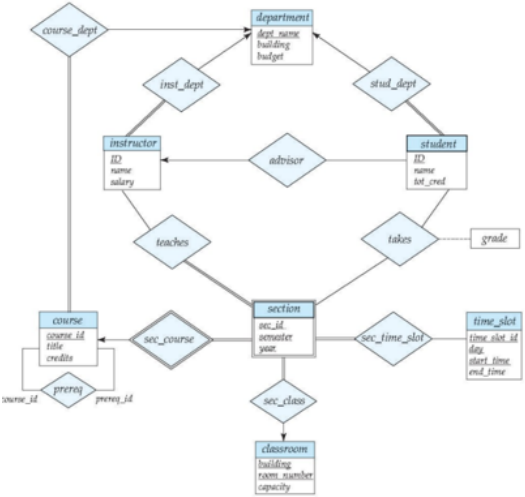
- Primary key for section - (course_id, sec_id, semester, year)

ER Diagram for a University Enterprise



Partha Pratim Das

ER Model to Relational Schema

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- Each schema has a number of columns (generally corresponding to attributes), which have unique names

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  - `section(course_id, sec_id, sem, year)`

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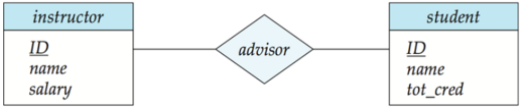

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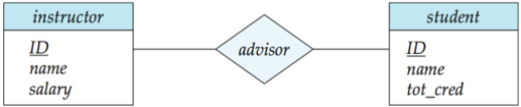

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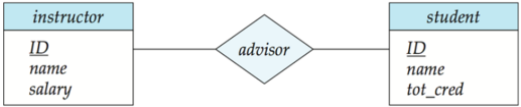

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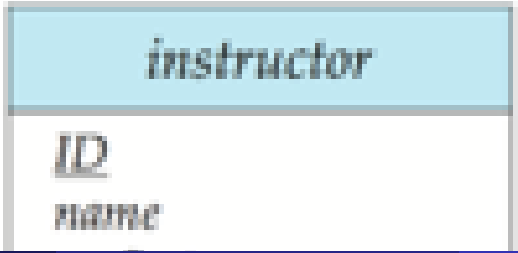

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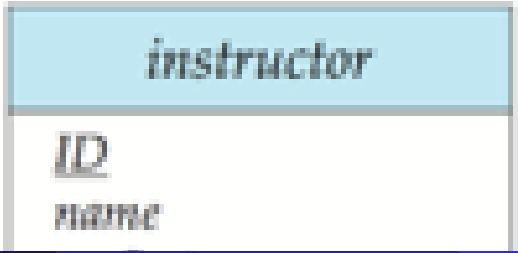

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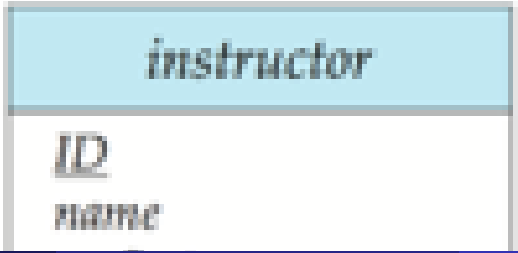

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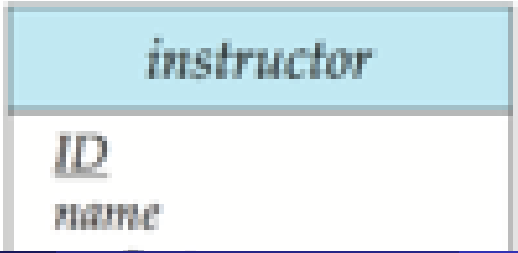

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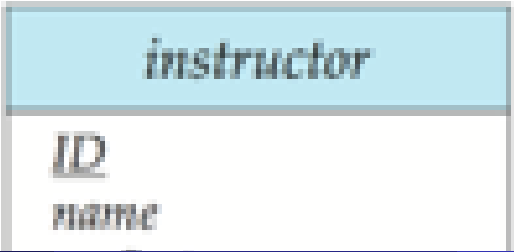
  - `instructor(ID, first_name, middle_initial, last_name, street_number, street_name, apt_number, city, state, zip_code, date_of_birth)`



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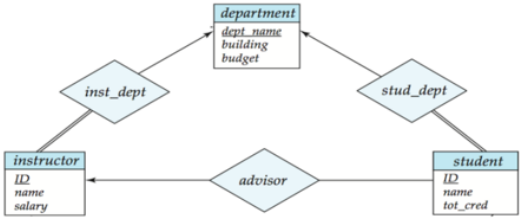
- Each value of the multivalued attribute maps to a separate tuple of the relation on schema E_M
- For example, an `instructor` entity with primary key 22222 and phone numbers 456-7890 and 123-4567 maps to two tuples: (22222, 456-7890) and (22222, 123-4567)

Redundancy of Schema

- Many-to-one and one-to-many relationship sets that are total on the many-side can be represented by adding an extra attribute to the 'many' side, containing the primary key of the 'one' side



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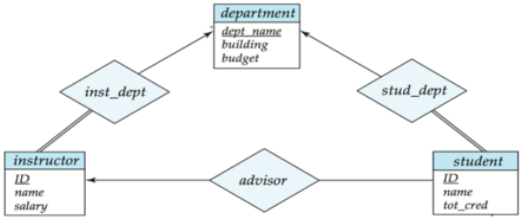
- Example: Instead of creating a schema for relationship set `inst_dept`, add an attribute `dept_name` to the schema arising from entity set `instructor`



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- For one-to-one relationship sets, either side can be chosen to act as the 'many' side
- That is, an extra attribute can be added to either of the tables corresponding to the two entity sets
- If participation is partial on the 'many' side, replacing a schema by an extra attribute in the schema corresponding to the 'many' side could result in null values

Redundancy of Schema (3)

- The schema corresponding to a relationship set linking a weak entity set to its identifying strong entity set is redundant.

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- Example: The `section` schema already contains the attributes that would appear in the `sec_course` schema

Module Summary

Summary

- Illustrated ER Diagram notation for ER Models

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- Illustrated ER Diagram notation for ER Models
- Discussed translation of ER Models to Relational Schema

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